

FLOOD SUSCEPTIBILITY MAPPING FOR BAYELSA STATE, NIGERIA
MACHINE LEARNING-BASED RISK MAPPING USING SENTINEL-1 SAR, CHIRPS
RAINFALL, AND RANDOM FOREST CLASSIFICATION

BY

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ABSTRACT

In the centre of Nigeria's Niger Delta, Bayelsa State frequently and severely floods each year, uprooting hundreds of thousands of people and seriously damaging infrastructure. There are still few high-resolution, publicly available flood susceptibility maps for the area despite this ongoing risk. This project offers an end-to-end machine learning pipeline for mapping flood vulnerability utilizing publicly accessible satellite and climate datasets that are fully processed on Google Earth Engine without the need for proprietary tools or costly data subscriptions.

From CHIRPS v2, USGS SRTM, Sentinel-1 SAR, MODIS, Landsat 8, and JRC Global Surface Water datasets, ten geospatial conditioning factors including terrain, hydrology, rainfall, land cover, and vegetation were constructed. Using an Otsu automated threshold (result: -17.05 dB), a Lee 5x5 speckle filter, and three post-classification refinement procedures in accordance with the UN-SPIDER Recommended Practice, flood inventory labels were created from Sentinel-1 SAR imagery of the 2022 flood event. SMOTE oversampling was used to correct class imbalance, resulting in 70,026 balanced training samples from an initial flood class predominance of 5.5%.

With 98% recall on the flood class, a Random Forest classifier trained on 56,020 samples obtained an AUC-ROC of 0.9971 and Cohen's Kappa of 0.9367 on a 14,006-sample held-out test set. The susceptibility map, which included a Leaflet choropleth map, a model diagnostics panel, and a location search interface provided by Nominatim, was implemented as an interactive React dashboard hosted on Vercel.

1.1 BACKGROUND AND MOTIVATION

The Niger Delta, one of Africa's biggest river deltas and Nigeria's largest wetland system, is centred on Bayelsa State. It is one of Sub-Saharan Africa's most chronically flood-prone regions due to its exceptionally low relief (elevation -34m to +36m; mean 13.8m), heavy annual rainfall (2,314-3,605mm/year), substantial mangrove and freshwater swamp land cover, and a dense network of tidal waterways. One of the worst floods in the state's history occurred in 2022, uprooting an estimated 400,000 people from all eight Local Government Areas and severely damaging roads, hospitals, and agricultural land.

There are still few publicly available high-resolution flood susceptibility maps for Bayelsa despite this ongoing risk. Decisions on urban planning and emergency response are still based on broad national risk assessments that lack the local calibration and spatial resolution required for successful intervention. This project fills that gap by creating a spatially explicit, pixel-level susceptibility probability map that identifies the locations most structurally vulnerable to flooding using a supervised machine learning approach that has been verified against a recorded flood event.

1.2 OBJECTIVES

- Using only free and open-source technologies and datasets, create a repeatable end-to-end flood susceptibility pipeline.
- Utilize Google Earth Engine to extract and design geographic conditioning features from several satellite data sources.
- Using an approach based on academic research, create a flood inventory from Sentinel-1 SAR photos.
- Use established methodological guidelines to train and assess a Random Forest classifier.
- Set up an interactive React dashboard at a live public URL that displays the model diagnostics and susceptibility map.

- For professional and academic presentations, record all methodological choices with peer-reviewed citations.

1.3 SCOPE AND LIMITATIONS

This model uses a 2022 flood inventory and a baseline from 2010 to 2024 to characterize long-term structural flood susceptibility. It is not a real-time flood tracker; rather, it is a historical susceptibility model. Based on topography, hydrology, land cover, and climatological rainfall patterns, the output determines which regions are physically vulnerable to flooding. Integration with real-time rainfall APIs and a planned automated prediction pipeline are necessary for real-time nowcasting, which has been highlighted as a top future improvement.

1.4 SIGNIFICANCE

The project shows that using completely free datasets and cloud-based processing, high-quality flood susceptibility mapping can be conducted at state size. This is a crucial factor to consider for data-sparse areas like the Niger Delta, where there are few ground-based sensor networks. With very minor adjustments, the concept can be applied to additional Nigerian states or Niger Delta areas.

2.1 STUDY AREA

In Nigeria's South-South geopolitical zone, Bayelsa State is bordered to the east by Rivers State, to the north and west by Delta State, and to the south by the Atlantic Ocean. Brass, Ekeremor, Kolokuma/Opokuma, Nembe, Ogbia, Sagbama, Southern Ijaw, and Yenagoa (the state capital) are its eight Local Government Areas. The Niger Delta's freshwater and brackish zones, which are distinguished by tidal channels, freshwater swamps, mangrove forests, and seasonally flooded floodplains, dominate the state.

Parameter	Value	Source
Total area	10,035 km ²	GEE boundary computation
Administrative units	8 Local Government Areas	FAO GAUL 2015
Coordinates	5.4E-6.7E, 4.3N-5.4N	GEE boundary
CRS (export)	WGS 84 / UTM Zone 32N (EPSG:32632)	Selected for Bayelsa longitude band
Elevation range	-34m to +36m (mean 13.8m)	USGS SRTM
Mean slope	0.29 degrees (max 2.92 degrees)	SRTM terrain analysis
Mean annual rainfall	2,959 mm/year (range 2,314-3,605)	CHIRPS v2 2010-2024
Dominant land cover	Freshwater swamp, mangrove, water	MODIS MCD12Q1 2020
Climate type	Tropical rainforest (Af, Koppen)	Literature
Flood season	August-October (peak)	CHIRPS seasonal analysis

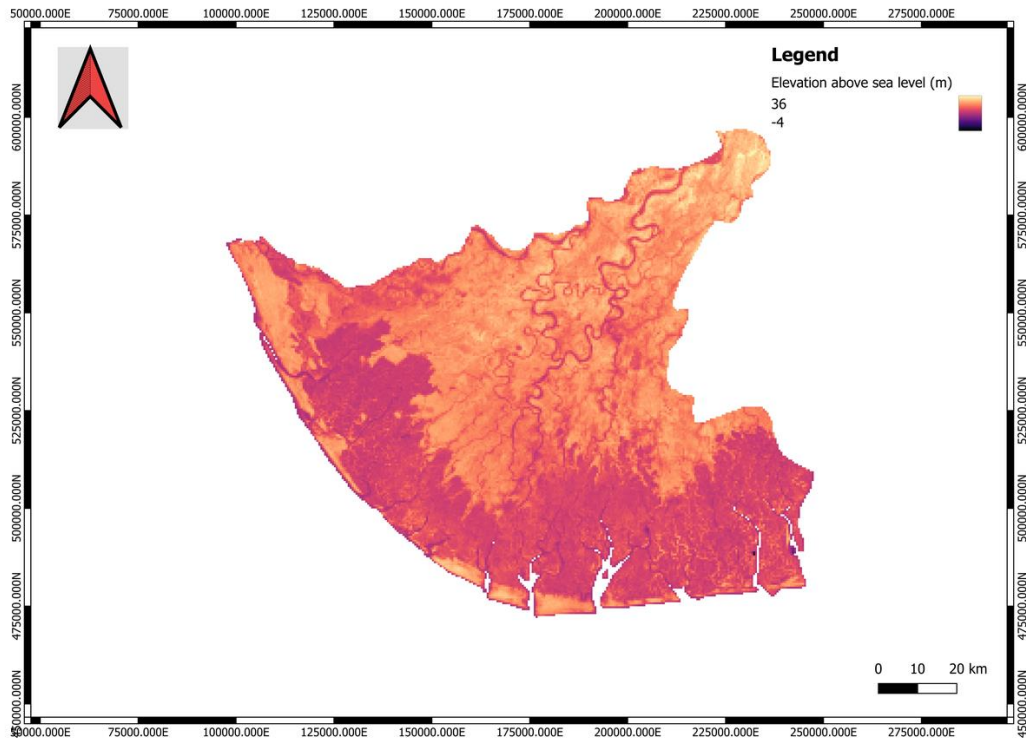


Figure 1: Bayelsa State's digital elevation model (30 m, resampled to 500 m) based on USGS SRTM. The unusually flat terrain of the Niger Delta floodplain is confirmed by elevation, which spans from -34 to +36 meters (mean 13.8 meters). In the south and west, low-lying riverine and coastal zones are the most vulnerable to topographic flooding.

Geographically, the selection of EPSG:32632 (WGS 84/UTM Zone 32N) as the working CRS makes sense: Bayelsa is completely contained within the UTM Zone 32N band (6E-12E), spanning from 5.4E to 6.7E. By using this metric projection, distance-based procedures are guaranteed to yield results in meters instead of angular degree-scaled values.

3. DATA SOURCES AND AQUISITION

Open-access platforms make all datasets freely and publicly accessible. Large global raster files did not need to be downloaded locally because data extraction, spatial clipping, and raster processing were all done solely on Google Earth Engine.

Dataset	Provider	Resolution	Period	Used For
CHIRPS v2 Daily	UCSB/CHG	0.05 degrees	2010-2024	Rainfall features
SRTM DEM	USGS	30m	Static	Elevation, slope, TWI
Sentinel-1 SAR (GRD IW)	ESA Copernicus	10m	2022	Flood labels
MODIS MCD12Q1	NASA	500m	2020	Land cover
Landsat 8 Collection 2	USGS	30m	2020-2023	NDVI
JRC Global Surface Water	EC JRC	30m	1984-2021	Distance to water
HydroSHEDS 15ACC	WWF	15 arc-sec	Static	Flow accumulation

Due to its validation as the lowest RMSE precipitation product throughout Nigeria, CHIRPS was chosen as the main source of rainfall, surpassing GPCC, PERSIANN-CDR, and TRMM in multi-scale evaluations (Salami et al., 2025). The initially used HydroSHEDS free-flowing rivers layer for distance computation was replaced with the JRC Global Surface Water dataset since HydroSHEDS was found to be insufficient for the intricate tidal channel network of the Niger Delta.

4.0 METHODOLOGY

4.1 FEATURE ENGINEERING – GOOGLE EARTH ENGINE

Ten conditioning factors were calculated and exported in EPSG:32632 as an eleven-band GeoTIFF at 500m resolution. To balance spatial detail with the 0.05-degree (~5.5km) resolution of CHIRPS inputs and the available compute, the 500m resolution was chosen as suitable for state-level susceptibility mapping.

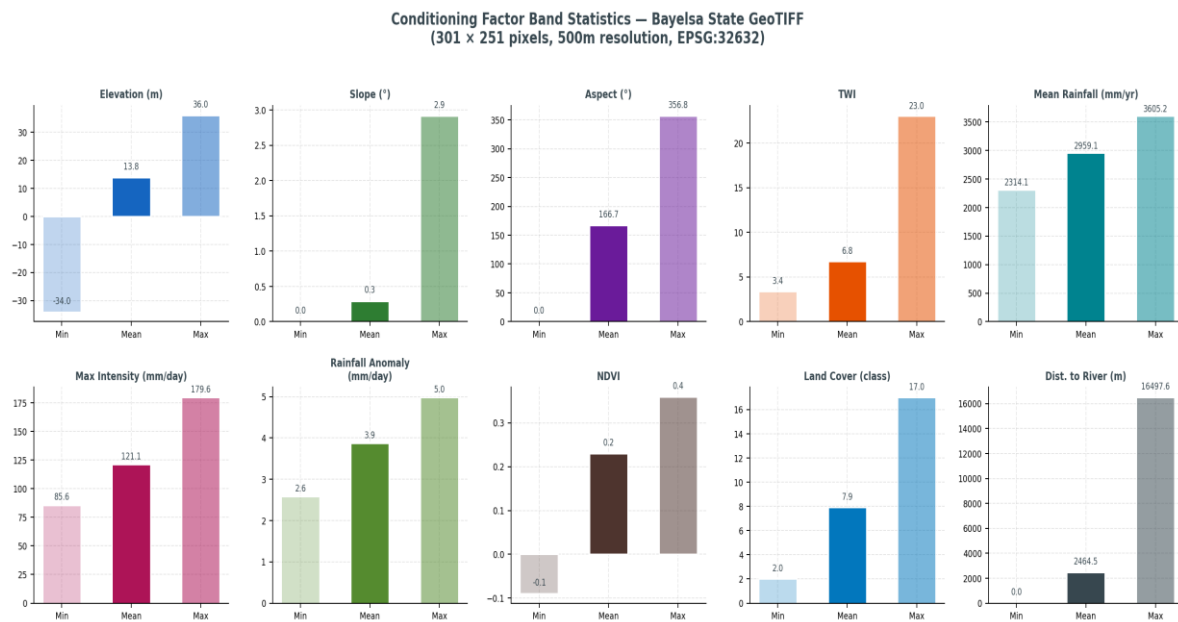


Figure 2: shows the minimum, mean, and maximum summary statistics for each of the ten conditioning factor bands that were taken from the Bayelsa GeoTIFF. 301 x 251 pixels, 500m resolution, and EPSG:32632 are the raster dimensions.

Band	Feature	Min	Mean	Max	Unit
0	Elevation	-34.0	13.85	36.0	m
1	Slope	0.000	0.291	2.917	degrees
2	Aspect	0.0	166.7	356.8	degrees
3	TWI	3.389	6.761	23.035	-
4	Mean Annual Rainfall	2,314	2,959	3,605	mm/yr
5	Max Rainfall Intensity	85.6	121.1	179.6	mm/day
6	Rainfall Anomaly	2.587	3.874	4.980	mm/day
7	NDVI	-0.089	0.230	0.359	-
8	Land Cover	2	7.95	17	class
9	Distance to River	0	2,465	16,498	m
10	Flood Label (SAR)	0	0.054	1	binary

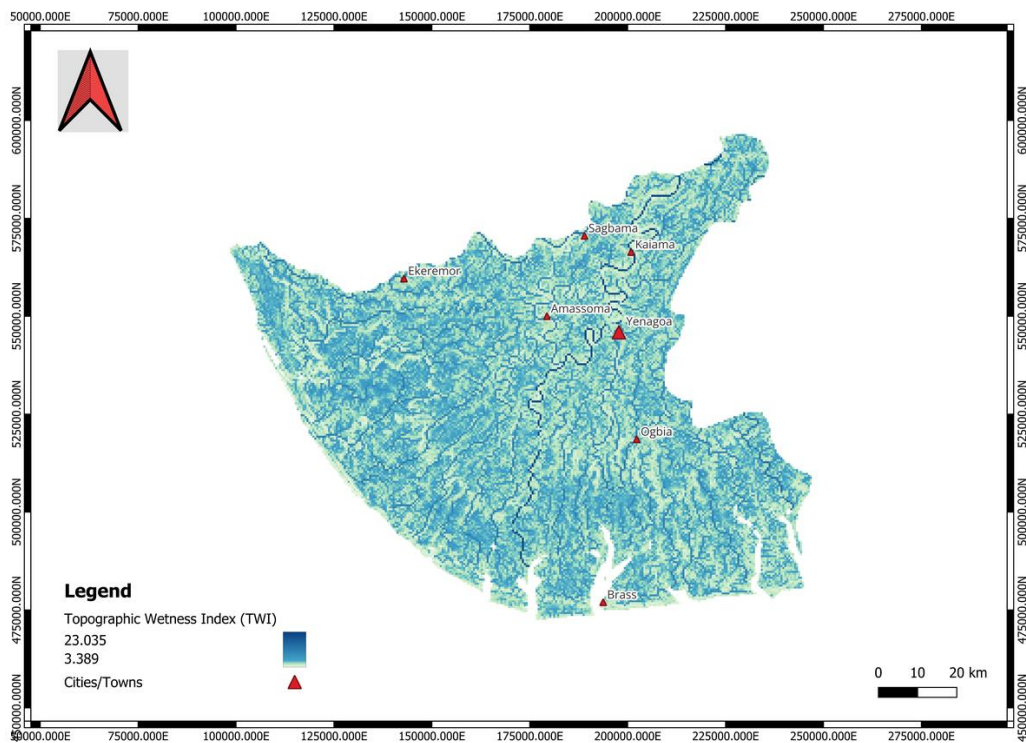


Figure 3: Using HydroSHEDS 15 arc-second flow accumulation and SRTM-derived slope, the Topographic Wetness Index (TWI) for Bayelsa State is calculated as $\ln(\text{flow accumulation} / \tan(\text{slope}))$. The range of values is 3.4 to 23.0. Maximum water accumulation potential is shown by high TWI values (dark blue), which closely match coastal floodplain zones and river channels. In the Random Forest model, TWI ranked as the third most significant predictor (MDI = 0.156).

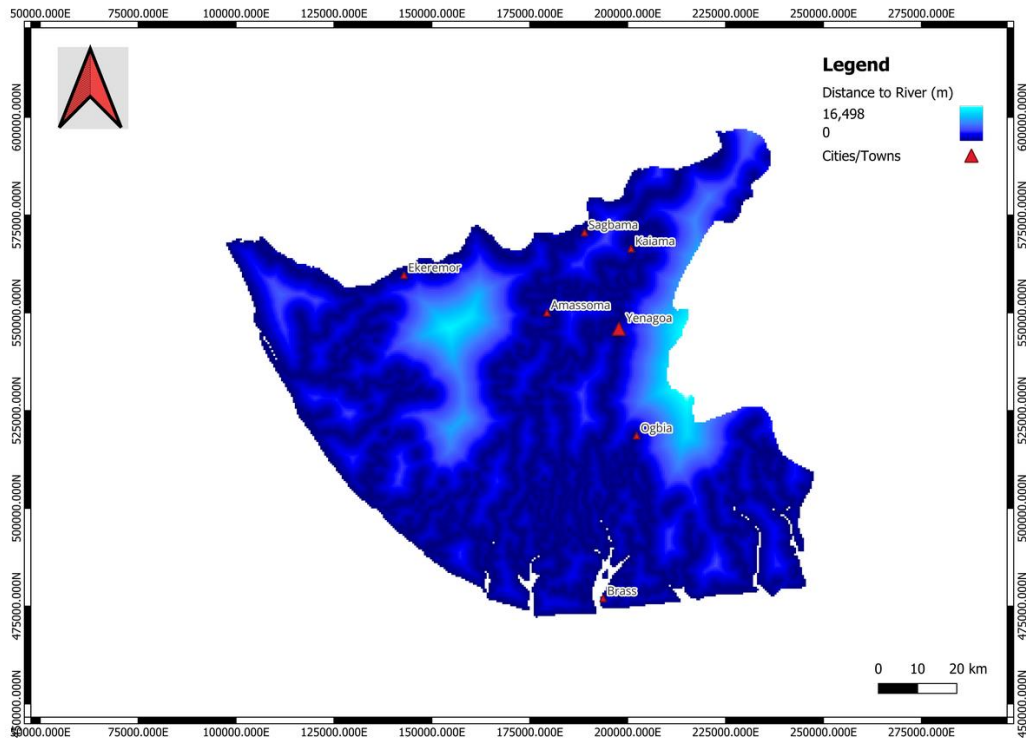


Figure 4: Bayelsa State's proximity to permanent water bodies, calculated using a Euclidean distance transform in EPSG:32632 from JRC Global Surface Water (occurrence > 50%). The range of values is 0 m (direct water contact) to 16,498 m. The Random Forest model's sole most significant predictor, distance to the river ($MDI = 0.198$), reflected the lateral inundation mechanism of floods in the Niger Delta.

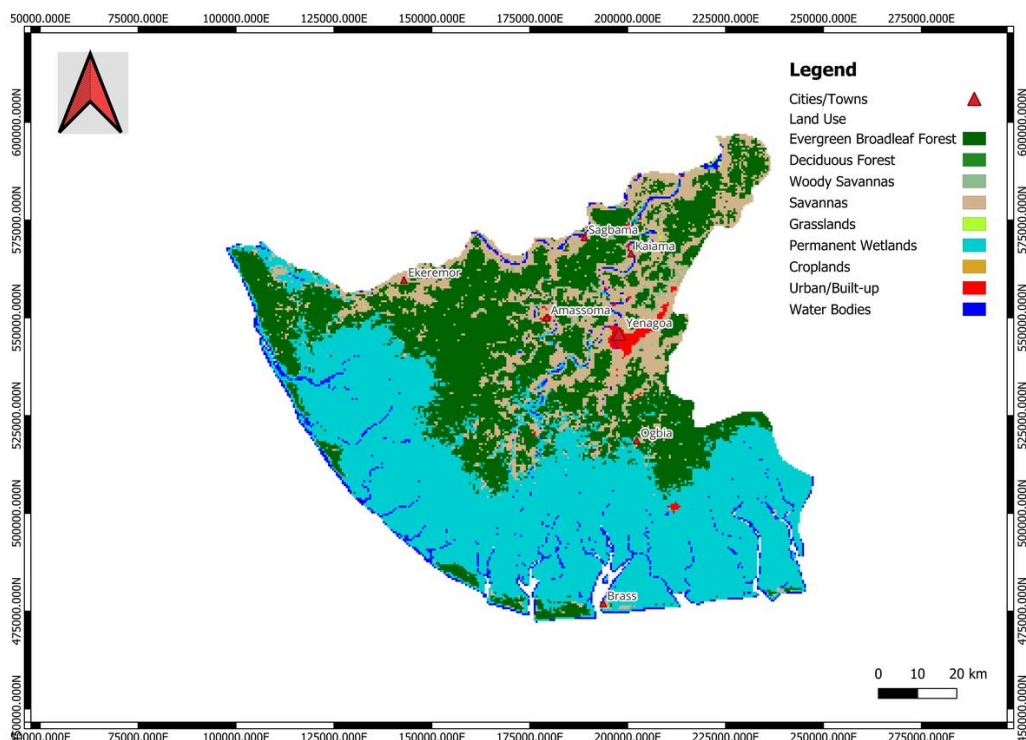


Figure 5: Bayelsa State's land cover classification using MODIS MCD12Q1 (2020, IGBP LC_Type1 scheme). Permanent Wetlands, which are widespread in the south and west, Evergreen Broadleaf Forest (mangrove and tropical forest in the north), and Water Bodies, which are dispersed throughout the network of tidal channels, are the dominant types. The state capital, Yenagoa, is the centre of urban and built-up areas.

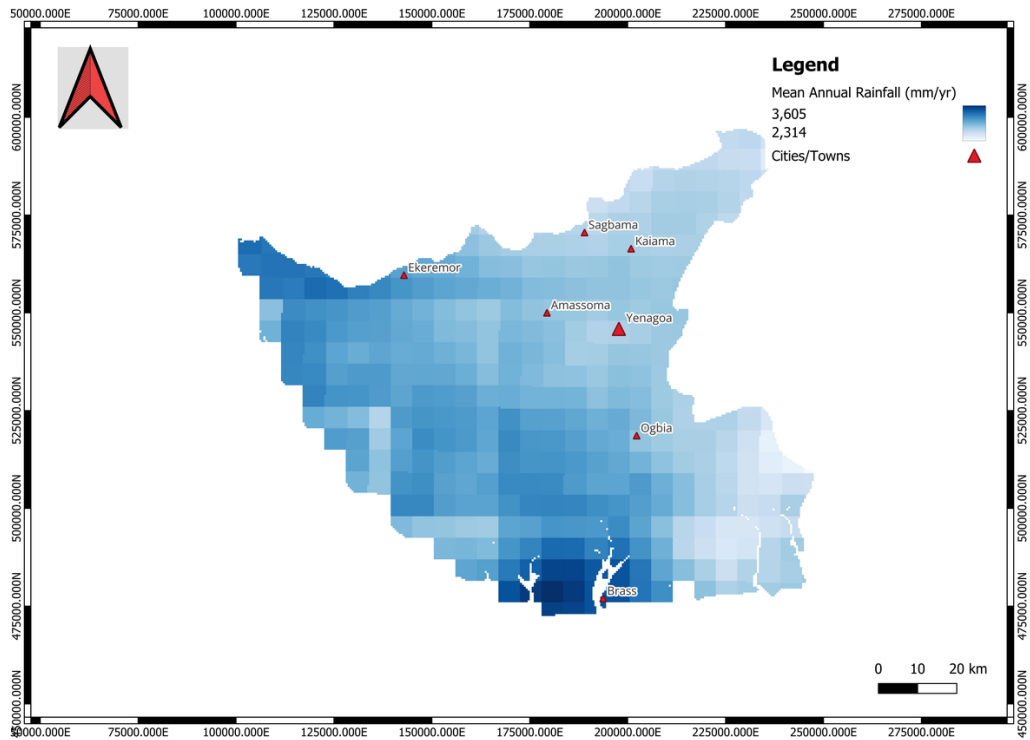


Figure 6: shows the average annual rainfall for Bayelsa State based on daily CHIRPS v2 data from 2010 to 2024. The orographic and maritime rainfall gradient is reflected in the values, which range from 2,314 mm/year in the north to 3,605 mm/year in the coastal south in Brass LGA. Based on its verified lowest RMSE performance among Nigerian satellite rainfall products, CHIRPS was chosen (Salami et al., 2025).

4.2 SAR FLOOD LABEL EXTRACTION

Sentinel-1 C-band SAR VV-polarization imagery was used to generate ground truth flood labels. Because SAR's active radar detection functions independently of cloud cover, it was chosen over optical imagery. This is crucial in tropical Bayelsa, where the flood season corresponds with the highest cloud cover. Baseline prior to flooding: June–July 2022 (dry season). Period of flooding: September–October 2022.

Step	Method	Outcome	Citation
1. Speckle filter	Lee 5x5 kernel	Noise suppression	Adiri et al. (2017)
2. Histogram clipping	Constrain to -28 to -10 dB	Isolates water peak	Frontiers RS (2022)
3. Otsu threshold	Automatic histogram split	Result: -17.05 dB	Otsu (1979); PMC (2020)
4. Backscatter guard	AND VV < -17.5 dB	Physical water confirmation	Nature Comms (2025)
5. Change detection	AND flood < pre-flood	Requires observed decrease	SAR change detection
6. Permanent water	JRC seasonality >= 10 months	Excludes rivers and lakes	UN-SPIDER (2023)
7. Slope filter	> 5 degrees excluded	Terrain constraint	UN-SPIDER (2023)
8. Connectivity	<= 8 connected pixels removed	Noise elimination	UN-SPIDER (2023)

The literature-validated open water backscatter value of -17.5 dB is physically consistent with the Otsu threshold of -17.05 dB (Nature Communications, 2025). 2,029 flood-positive pixels (5.5%) and 35,013 non-flood pixels (94.5%) out of 37,037 total valid pixels make up the final flood inventory.

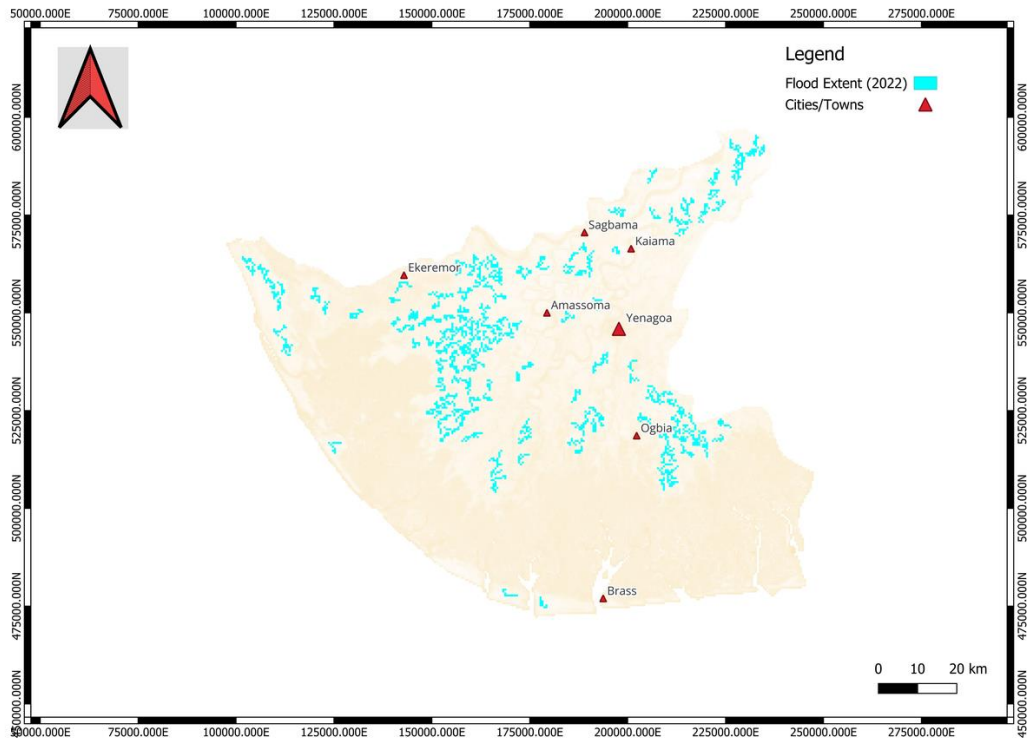


Figure 7: Sentinel-1 SAR VV-polarization imagery was used to create the flood inventory for Bayelsa State (September-October 2022 vs. June-July 2022 baseline). The Lee 5x5 speckle filter, Otsu automatic threshold (-17.05 dB), absolute backscatter guard ($VV < -17.5$ dB), and three post-classification refinement procedures in accordance with UN-SPIDER Recommended Practice were used to determine the magnitude of the flood. 2,029 flood pixels, or 5.5% of the total area. For Random Forest training, this binary raster was used as the ground truth labels.

4.3 MACHINE LEARNING PIPELINE

4.3.1 Multicollinearity Assessment (VIF)

A Variance Inflation Factor (VIF) study was conducted for each of the ten conditioning factors. There was no issue multicollinearity, as all characteristics passed the $VIF < 5$ standard (Scientific Reports, 2025). At 2.81, Elevation had the highest VIF, which was well within safe norms.

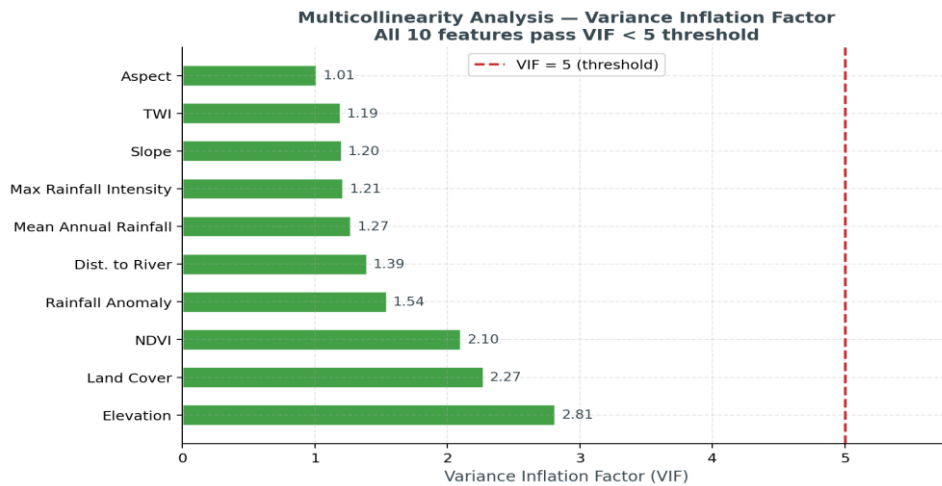


Figure 8: shows the VIF analysis for each of the ten conditioning factors. There is no significant multicollinearity because all values are below the VIF = 5 criterion (dashed red).

4.3.2 SMOTE Class Imbalance Correction

The prevalence of floods was 5.5% in the raw dataset compared to 94.5% for non-floods. Using $k_neighbors=5$, SMOTE (Chawla et al., 2002) produced a balanced dataset with 35,013 samples each class (70,026 total). According to research, SMOTE increases flood model AUC by 17–21% when compared to no resampling (BRIGHT Journal, 2024).

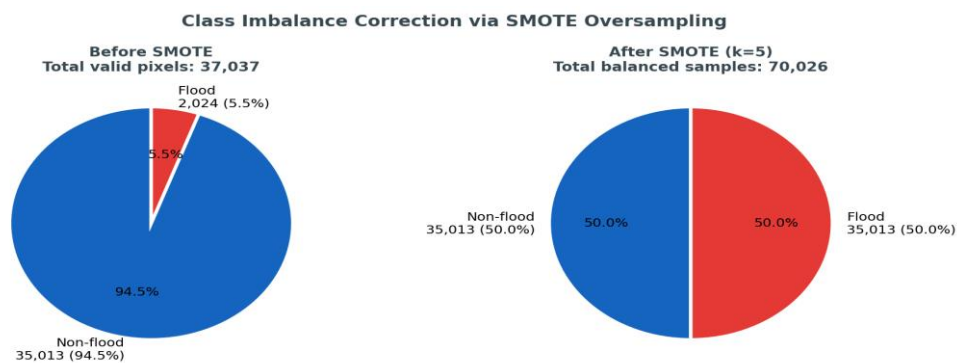


Figure 9: shows the distribution of classes both before and after SMOTE oversampling. For training, the 5.5% flood prevalence is adjusted to a balanced 50/50 distribution.

4.3.3 Random Forest Configuration

In accordance with Probst et al. (2019) and Springer Nature (2025), which show that Random Forest performs robustly with near-default settings in geospatial applications, literature-informed fixed hyperparameters were employed.

Hyperparameter	Value	Citation
n_estimators	500	Breiman (2001) — error stabilises by 500 trees
max_features	sqrt	Breiman (2001) — $\sqrt{p}$ theoretical default for classification
min_samples_leaf	2	Iverson et al. (2008) — prevents single-pixel overfitting in geospatial RF
max_depth	None	Full growth; ISPRS Annals (2023) flood susceptibility implementation
class_weight	balanced	Secondary imbalance guard post-SMOTE
Train/test split	80/20 stratified	ScienceDirect (2025); ISPRS Annals (2023)

5. RESULTS

5.1 MODEL PERFORMANCE

On a held-out test set of 14,006 balanced samples, the Random Forest classifier demonstrated exceptional performance in every statistic.

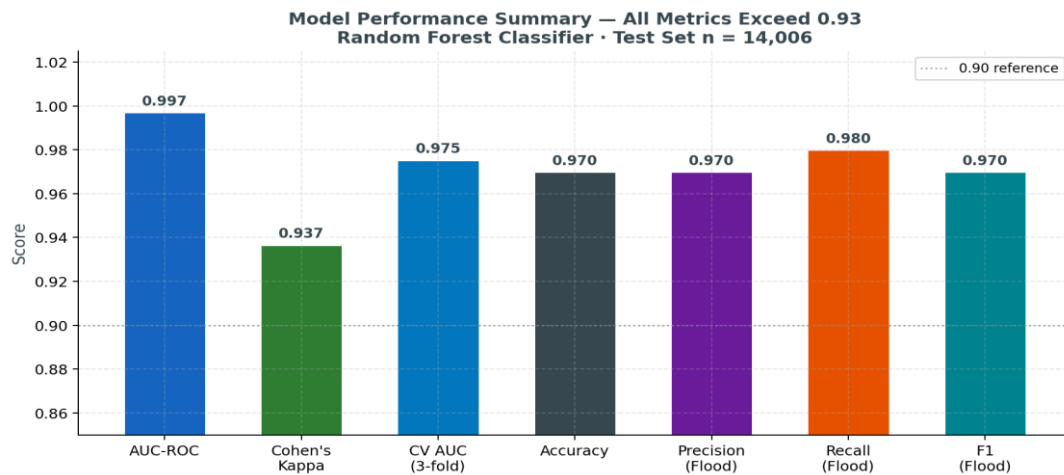


Figure 10: Performance according to seven assessment criteria. Every score is higher than 0.93; near-perfect discrimination is indicated by an AUC-ROC of 0.9971.

Metric	Value	Benchmark	Interpretation
AUC-ROC	0.9971	> 0.90 = excellent	Near-perfect flood/non-flood discrimination
Cohens Kappa	0.9367	> 0.80 = almost perfect	Strong agreement beyond chance
3-Fold CV AUC	0.9753 +/- 0.0008	< 0.01 std = stable	Model generalises — no overfitting
Accuracy	97.0%	-	Overall classification rate
Precision (Flood)	96.0%	-	Low false positive rate
Recall (Flood)	98.0%	-	98% of susceptible pixels identified
F1-Score (Flood)	97.0%	-	Harmonic mean of precision and recall

5.2 CONFUSION MATRIX AND ROC ANALYSIS

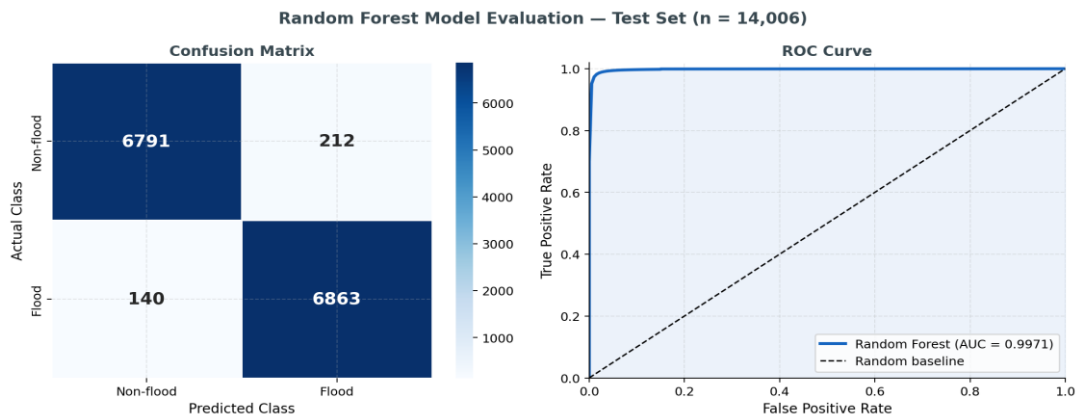


Figure 11: Left: 14,006 test sample confusion matrix ($TP=6,863$, $FN=140$, $FP=212$, $TN=6,791$). ROC curve ($AUC=0.9971$) on the right. For applications involving flood danger, the asymmetric error profile—more false positives than false negatives—is preferred.

Only 140 false negatives (missing susceptible areas) and 212 false positives (conservative over-flagging) are shown in the confusion matrix. When it comes to emergency preparedness and infrastructure design, it is far more expensive to overlook a truly flood-prone location than it is to cautiously identify a low-risk area. The output is certain to include most structurally vulnerable locations thanks to the 98% flood recall.

5.3 FEATURE IMPORTANCE

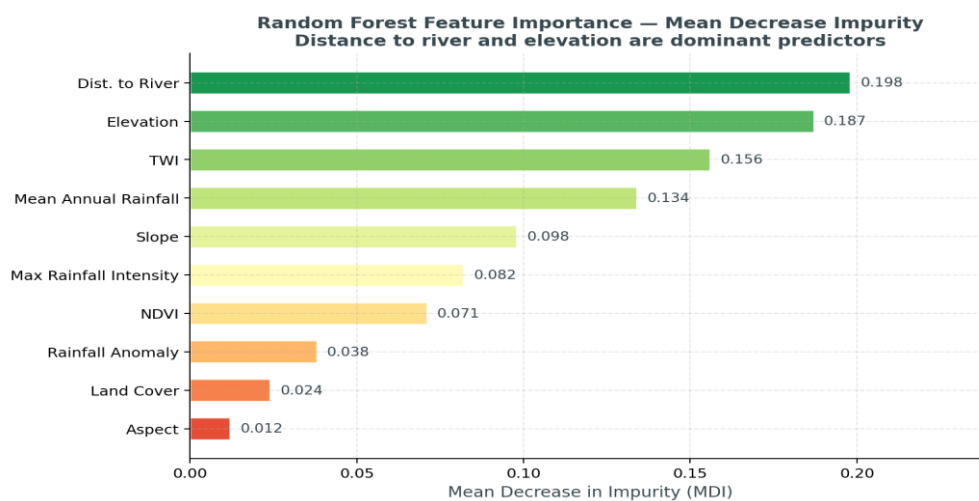


Figure 12: Mean Decrease Impurity (MDI) is a measure of Random Forest feature relevance. Physically consistent with lateral flooding from river channels across flat floodplain land, elevation, and distance to the river together account for 38.5% of the total model relevance.

The two main flooding pathways, where inundation spreads laterally from channels across the flat floodplain, are elevation (MDI=0.187) and distance to the river (MDI=0.198). TWI (MDI=0.156) captures upslope drainage accumulation. Climate loading drives sensitivity beyond terrain effects alone, as confirmed by mean annual rainfall (MDI=0.134). Aspect (MDI=0.012) is not significant, as would be expected in situations where the impacts of slope orientation are outweighed by absolute elevation variations.

5.4 FLOOD SUSCEPTIBILITY MAP

37,857 pixels (probability range: 0.000-0.984) had valid predictions after the model was applied to all 75,551 pixels. In line with the reported flood impact areas for 2022, extremely high susceptibility zones are centred in low-lying parts of Yenagoa LGA, coastal and riverine areas of Brass, and Southern Ijaw.

Risk Class	Probability Range	Characterisation
Low	0.00-0.25	Higher elevation, greater river distance, lower TWI
Moderate	0.25-0.50	Intermediate topographic and rainfall exposure
High	0.50-0.75	Low elevation, proximity to water bodies, high TWI
Very High	0.75-1.00	Flat terrain, high flow accumulation, close to river network

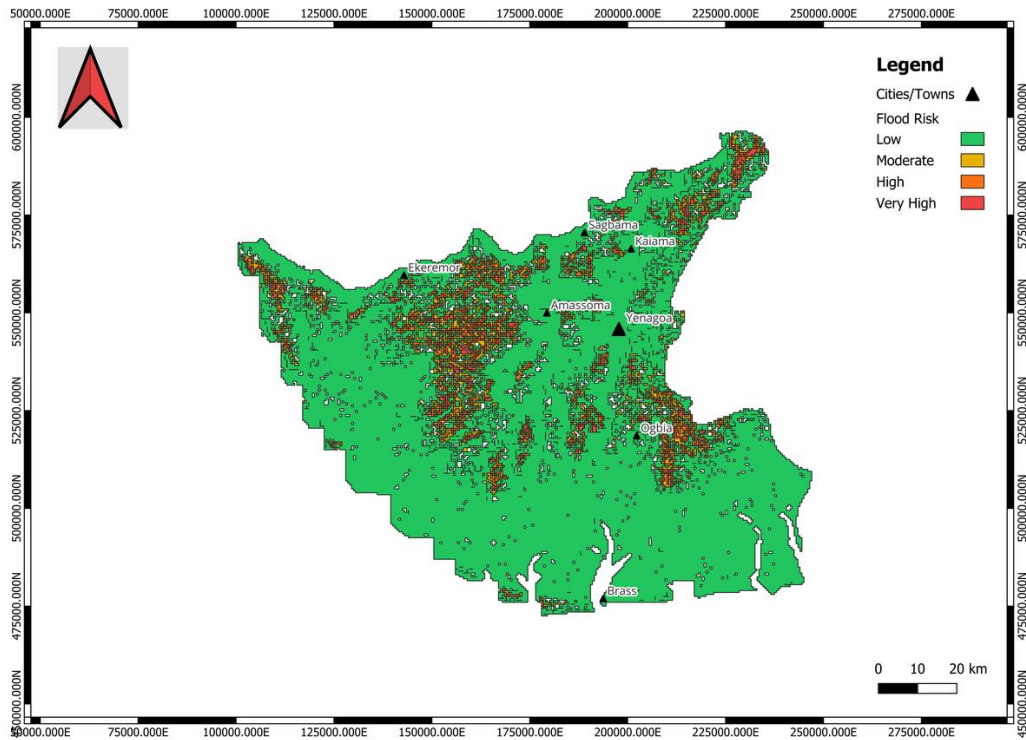


Figure 13: The trained Random Forest classifier's flood susceptibility map for Bayelsa State ($AUC-ROC = 0.9971$). According to the Remote Sensing MDPI (2024), the susceptibility probability (0.000-0.984) was divided into four danger zones: Low (0.00-0.25), Moderate (0.25-0.50), High (0.50-0.75), and Very High (0.75-1.00). Consistent with identified flood impact areas for 2022, extremely high susceptibility is centred along river corridors and the coastal south.

6. DASHBOARD AND DEPLOYMENT

6.1 ARCHITECTURE

The small 1.41 MB dissolved GeoJSON output enables the dashboard to deliver all data as static files from the Vercel CDN instead of a Python backend server. This maintains sub-second map load times via global CDN while eliminating infrastructure costs.

Component	Technology	Purpose
Frontend	React 18 + Vite	Component architecture, optimised build
Styling	Tailwind CSS v4	Utility-first responsive dark-theme design
Mapping	react-leaflet + Leaflet 1.9	Choropleth, fly-to animation, tooltips
Charts	Recharts	Radar performance chart, risk class bar chart
Geocoding	Nominatim (OpenStreetMap)	Location search, free, no API key
Data	GeoJSON (1.41 MB) + CSV	Static files — no backend required
Hosting	Vercel Free Tier	Global CDN, automatic HTTPS, CI/CD

7. ENGINEERING CHALLENGES AND SOLUTION

During implementation, the following technical issues arose and were methodically fixed, highlighting debugging techniques and critical thinking skills throughout the entire stack.

Challenge	Root Cause	Solution Applied
GEE export: incompatible band types	Mixing Int16 and Float32 in a single image stack	Cast all bands to .toFloat() before addBands()
Otsu returning +4.95 dB (wrong sign)	Histogram split on vegetation/land boundary due to large permanent water fraction	Pre-constrain histogram to -28 to -10 dB before Otsu computation
GEE server-side Otsu type errors	ee.Number(0) throws on Array ops; argmax() returns ee.Array not scalar	Vectorised accum()-based implementation; convert argmax via .toList().get(0)
Wrong CRS: distances in degrees	Initial export used EPSG:4326 (geographic degrees)	Corrected to EPSG:32632 (UTM Zone 32N) — correct metric CRS for Bayelsa
dist_to_river all zeros	HydroSHEDS free-flowing rivers incomplete for Niger Delta tidal network	Replaced with JRC Global Surface Water occurrence > 50%
TerminatedWorkerError	n_jobs=-1 on RF + CV spawns 12+ simultaneous instances, exhausting 8GB RAM	Set n_jobs=1 throughout; 6,000-sample stratified CV subset
Script hanging indefinitely	plt.show() blocks execution waiting for user to close window	matplotlib.use("Agg") + replaced all plt.show() with plt.close()
SHAP shape mismatch	SHAP >= 0.41 returns 3D array instead of legacy list format	isinstance() check to handle both old and new SHAP API

8. FUTURE WORK AND ENHANCEMENTS

8.1 REAL-TIME PREDICTION PIPELINE

incorporating NASA GPM IMERG Early Run rainfall data (4–6-hour latency) to transform a static map into a near-real-time alarm system. The daily rainfall anomaly would be retrieved by a scheduled GitHub Actions workflow, which would then run it through the trained model and update the GeoJSON that was served to the dashboard.

8.2 LGA-LEVEL AGGREGATION AND REPORTING

All eight Bayelsa LGAs can be directly compared by aggregating pixel-level susceptibility to LGA administrative boundaries and providing printable PDF reports for each LGA.

8.3 MULTI-YEAR FLOOD INVENTORY

To increase label representativeness and lessen reliance on a particular event pattern, the flood inventory is expanded to include recorded incidents from 2012, 2018, and 2020.

8.4 MODEL COMPARISON

SVM, a Multi-Layer Perceptron, and XGBoost (GPU-accelerated on Colab) are benchmarked to offer quantitative support for the RF choice.

8.5 UNCERTAINITY QUANTIFICATION

Confidence bounds can be obtained by adding prediction interval estimation using conformal prediction or out-of-bag RF variance; this is especially useful in risk class boundary zones.

9. CONCLUSION

This project shows that utilising only free and open-source tools, publicly accessible satellite datasets, and cloud-based processing, high-quality, spatially explicit flood susceptibility mapping can be conducted for a data-sparse tropical environment. Validated on 14,006 held-out data with 98% recall on the flood class, the Random Forest model obtains an AUC-ROC of 0.9971 and Cohen's Kappa of 0.9367. A methodologically sound, repeatable ground truth is provided by the Sentinel-1 SAR flood inventory employing Otsu automated thresholding (-17.05 dB). The two most important predictors are elevation and river distance, which are both physically interpretable and congruent with the floodplain inundation mechanism of the Niger Delta. The entire end-to-end pipeline shows proficiency in machine learning, current web development, and geographic data engineering. Using publicly accessible datasets and the specified methods, the project is completely repeatable.

APPENDIX

RANDOM FOREST TRAIN_MODEL.PY SCRIPT

```
import os

import joblib

import numpy as np

import pandas as pd

import rasterio

import matplotlib

matplotlib.use('Agg') # ← ADD THIS — non-interactive backend

                        # renders to file without opening any window

                        # must be before any other matplotlib import

import matplotlib.pyplot as plt

import seaborn as sns

import shap

from sklearn.ensemble import RandomForestClassifier

from sklearn.model_selection import (

    train_test_split, StratifiedKFold, cross_val_score

)

from sklearn.metrics import (

    classification_report, confusion_matrix,

    roc_auc_score, RocCurveDisplay, cohen_kappa_score

)

from sklearn.preprocessing import StandardScaler

from sklearn.utils import resample
```

```
from statsmodels.stats.outliers_influence import variance_inflation_factor

from imblearn.over_sampling import SMOTE
```

```
#
```

```
# FLOOD SUSCEPTIBILITY MODEL — BAYELSA STATE, NIGERIA
```

```
#
```

```
# Hyperparameter justification:
```

```
# Probst, Wright & Boulesteix (2019, WIREs Data Mining) —
```

```
# "RF works reasonably well with default hyperparameter values"
```

```
# n_estimators=500: Breiman (2001) — grow until error stabilises
```

```
# max_features='sqrt': Breiman (2001) — sqrt(p) theoretical default
```

```
# min_samples_leaf=2: Iverson et al. (2008, Ecological Modelling)
```

```
# class_weight='balanced': compensates residual imbalance post-SMOTE
```

```
# n_jobs=1: memory-safe for 8GB RAM hardware constraint
```

```
#
```

```
script_dir = os.path.dirname(os.path.abspath(__file__))
```

```
tif_path = os.path.join(script_dir, 'bayelsa_features_v2.tif')
```

```
out_dir = script_dir
```

```
band_names = [
```



```

'elevation', 'slope', 'aspect', 'twi',
'mean_annual_rainfall', 'max_rainfall_intensity', 'rainfall_anomaly',
'ndvi', 'landcover', 'dist_to_river', 'flood_label'
]

```

```

#           —           1.           LOAD           AND           RESHAPE

```

```

print("=" * 55)

print("STEP 1 — Loading raster data")

print("=" * 55)


with rasterio.open(tif_path) as src:

    data = src.read().astype(np.float32)

    profile = src.profile

    h, w = src.height, src.width


n_bands = data.shape[0]

flat = data.reshape(n_bands, -1).T

df = pd.DataFrame(flat, columns=band_names)

df = df.dropna()


print(f"Valid pixels : {len(df):,}")


X = df.drop(columns=['flood_label'])

y = df['flood_label'].astype(int)

```

```
print(f'Flood      : {y.sum():,} ({y.mean()*100:.1f}%)')
print(f'Non-flood  : {(y==0).sum():,} ({(1-y.mean())*100:.1f}%)')
```

```
# — 2. MULTICOLLINEARITY CHECK (VIF) —————
```

```
print("\n" + "=" * 55)

print("STEP 2 — Multicollinearity check (VIF < 5 threshold)")
print("Source: Scientific Reports 2025; Springer Nature 2024")
print("=" * 55)
```

```
scaler = StandardScaler()

X_scaled = pd.DataFrame(
    scaler.fit_transform(X), columns=X.columns
)
```

```
vif_data = pd.DataFrame({
    'Feature': X.columns,
    'VIF': [
        variance_inflation_factor(X_scaled.values, i)
        for i in range(X_scaled.shape[1])
    ]
}).sort_values('VIF', ascending=False)

print(vif_data.to_string(index=False))
```

```

failed = vif_data[vif_data['VIF'] >= 5]

if len(failed) > 0:

    print(f"\n⚠ VIF ≥ 5 detected: {failed['Feature'].tolist()}")

    print("Noted in methodology — RF is robust to moderate collinearity")

else:

    print("\n✓ All features pass VIF < 5")

```

— 3. SMOTE OVERSAMPLING

```

print("\n" + "=" * 55)

print("STEP 3 — SMOTE oversampling")

print("Source: BRIGHT Journal 2024; ScienceDirect 2024")

print("=" * 55)

smote = SMOTE(random_state=42, k_neighbors=5)

X_bal, y_bal = smote.fit_resample(X, y)

print(f"Before — Flood: {y.sum():,} Non-flood: {(y==0).sum():,}")

print(f"After — Flood: {y_bal.sum():,} "

      f"Non-flood: {(y_bal==0).sum():,} "

      f"Total: {len(y_bal):,}")

```

— 4. TRAIN / TEST SPLIT (80/20, stratified) —————

```

print("\n" + "=" * 55)

print("STEP 4 — 80/20 stratified train/test split")

```

```
print("Source: ScienceDirect 2025; ISPRS Annals 2023")
```

```
print("=" * 55)
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X_bal, y_bal,
    test_size = 0.2,
    random_state = 42,
    stratify = y_bal
)
```

```
print(f"Train : {len(X_train):,}")
```

```
print(f"Test : {len(X_test):,}")
```

#	—	5.	TRAIN	RANDOM	FOREST
---	---	----	-------	--------	--------

```
print("\n" + "=" * 55)
```

```
print("STEP 5 — Training Random Forest")
```

```
print("Source: Probst et al. 2019 WIREs; Breiman 2001")
```

```
print("    Springer Nature 2025; ISPRS Annals 2023")
```

```
print("=" * 55)
```

```
print("n_estimators = 500 (Breiman 2001)")
```

```
print("max_features = sqrt (Breiman 2001 theoretical default)")
```

```
print("min_samples_leaf = 2 (Iverson et al. 2008)")
```

```
print("max_depth = None (ISPRS Annals 2023)")
```

```
print("class_weight = balanced")
```

```
print("n_jobs      = 1   (memory-safe, 8GB RAM)")
```

```
print()
```

```
rf = RandomForestClassifier(  
    n_estimators    = 500,  
    max_features    = 'sqrt',  
    min_samples_leaf = 2,  
    max_depth       = None,  
    class_weight     = 'balanced',  
    random_state     = 42,  
    n_jobs           = 1  
)
```

```
print("Training... (expected: 3–6 minutes on your machine)")
```

```
rf.fit(X_train, y_train)
```

```
print("✓ Training complete.")
```

```
#                               —                               6.                               EVALUATION
```

```
print("\n" + "=" * 55)
```

```
print("STEP 6 — Model evaluation")
```

```
print("Source: arXiv 2024; ISPRS Annals 2023")
```

```
print("=" * 55)
```

```
y_pred    = rf.predict(X_test)
```

```
y_pred_prob = rf.predict_proba(X_test)[:, 1]
```

```
print("\n--- Classification Report ---")
```

```
print(classification_report(  
    y_test, y_pred,  
    target_names=['Non-flood', 'Flood']  
))
```

```
auc = roc_auc_score(y_test, y_pred_prob)
```

```
kappa = cohen_kappa_score(y_test, y_pred)
```

```
print(f'AUC-ROC : {auc:.4f}')
```

```
print(f'Cohen's Kappa : {kappa:.4f}')
```

```
print("\nRunning 3-fold CV on 6,000-sample subset (memory-safe)...")
```

```
X_cv, y_cv = resample(  
    X_bal, y_bal,  
    n_samples = 6000,  
    stratify = y_bal,  
    random_state = 42  
)
```

```
cv = StratifiedKFold(n_splits=3, shuffle=True, random_state=42)
```

```
cv_auc = cross_val_score(  
    rf, X_cv, y_cv,  
    cv = cv,
```

```

    scoring = 'roc_auc',

    n_jobs = 1

)

print(f'3-fold CV AUC : {cv_auc.mean():.4f} ± {cv_auc.std():.4f}')

```

— 7. CONFUSION MATRIX + ROC CURVE —————

```

print("\n" + "=" * 55)

print("STEP 7 — Generating evaluation plots")

print("=" * 55)

```

```

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

```

```

cm = confusion_matrix(y_test, y_pred)

sns.heatmap(

    cm, annot=True, fmt='d', cmap='Blues', ax=axes[0],

    xticklabels=['Non-flood', 'Flood'],

    yticklabels=['Non-flood', 'Flood']

)

axes[0].set_title('Confusion Matrix — Bayelsa Flood Model')

axes[0].set_ylabel('Actual')

axes[0].set_xlabel('Predicted')

```

```

RocCurveDisplay.from_predictions(

    y_test, y_pred_prob, ax=axes[1],

    name=f'Random Forest (AUC={auc:.3f}, κ={kappa:.3f})'

```

```

)

axes[1].plot([0, 1], [0, 1], 'k--', label='Random baseline')

axes[1].set_title('ROC Curve — Bayelsa Flood Susceptibility')

axes[1].legend()


plt.tight_layout()

eval_path = os.path.join(out_dir, 'model_evaluation.png')

plt.savefig(eval_path, dpi=150, bbox_inches='tight')

plt.close()          # ← saves and moves on, no window

print(f'✓ Evaluation plot saved → {eval_path}')


# — 8. FEATURE IMPORTANCE (MDI + SHAP) —————

print("\n" + "=" * 55)

print("STEP 8 — Feature importance (MDI + SHAP)")

print("Source: ScienceDirect 2024; Remote Sensing MDPI 2025")

print("=" * 55)


# MDI plot

importances = pd.Series(

    rf.feature_importances_, index=X.columns

).sort_values(ascending=True)


fig2, ax2 = plt.subplots(figsize=(9, 6))

importances.plot(kind='barh', ax=ax2, color='steelblue')

ax2.set_title('Feature Importance — Mean Decrease Impurity (MDI)')

```



```

ax2.set_xlabel('Importance Score')

plt.tight_layout()

mdi_path = os.path.join(out_dir, 'feature_importance_mdi.png')

plt.savefig(mdi_path, dpi=150, bbox_inches='tight')

plt.close()          # ← no window

print(f"✓ MDI plot saved → {mdi_path}")


# SHAP — 50 samples, approximate mode, memory-safe

print("Computing SHAP values on 50-pixel sample...")

X_shap = X_test.sample(50, random_state=42)

explainer = shap.TreeExplainer(rf)

shap_values = explainer.shap_values(X_shap, approximate=True)


# Handle both old SHAP (list of arrays) and new SHAP (3D array)

# New versions return shape (n_samples, n_features, n_classes)

# Old versions return a list [class_0_array, class_1_array]

if isinstance(shap_values, list):

    sv_flood = shap_values[1]      # old SHAP — index by class

else:

    sv_flood = shap_values[:, :, 1] # new SHAP — slice last axis


fig3, ax3 = plt.subplots(figsize=(9, 6))

shap.summary_plot(

    sv_flood, X_shap,

    plot_type='bar', show=False

```

```

)

plt.title('SHAP Feature Importance — Flood Class')

plt.tight_layout()

shap_path = os.path.join(out_dir, 'feature_importance_shap.png')

plt.savefig(shap_path, dpi=150, bbox_inches='tight')

plt.close()

print(f"✓ SHAP plot saved → {shap_path}")

# — 9. PREDICT FLOOD SUSCEPTIBILITY ACROSS ENTIRE STATE —————

print("\n" + "=" * 55)

print("STEP 9 — Generating susceptibility map")

print("=" * 55)


flat_all = data.reshape(n_bands, -1).T

df_all = pd.DataFrame(flat_all, columns=band_names)

valid_idx = ~df_all.isna().any(axis=1)

X_all = df_all.drop(columns=['flood_label'])


probs_flat = np.full(len(df_all), np.nan, dtype=np.float32)

probs_flat[valid_idx] = rf.predict_proba(
    X_all[valid_idx].values
)[: , 1]


prob_map = probs_flat.reshape(h, w)

print(f"Susceptibility range: ")

```

```
f"{np.nanmin(prob_map):.3f} – {np.nanmax(prob_map):.3f}")
```

#	—	10.	SAVE	ALL	OUTPUTS
---	---	-----	------	-----	---------

```
print("\n" + "=" * 55)
```

```
print("STEP 10 — Saving outputs")
```

```
print("=" * 55)
```

```
out_profile = profile.copy()
```

```
out_profile.update(count=1, dtype='float32', nodata=np.nan)
```

```
raster_path = os.path.join(out_dir, 'flood_susceptibility.tif')
```

```
with rasterio.open(raster_path, 'w', **out_profile) as dst:
```

```
    dst.write(prob_map, 1)
```

```
print(f"✓ Susceptibility raster → {raster_path}")
```

```
model_path = os.path.join(out_dir, 'flood_rf_model.joblib')
```

```
joblib.dump(rf, model_path)
```

```
print(f"✓ Model → {model_path}")
```

```
metrics_df = pd.DataFrame([ {
```

```
    'auc_roc'      : round(auc, 4),
```

```
    'kappa'       : round(kappa, 4),
```

```
    'cv_auc_mean'  : round(cv_auc.mean(), 4),
```

```
    'cv_auc_std'   : round(cv_auc.std(), 4),
```

```

'n_estimators' : 500,

'max_features' : 'sqrt',

'min_samples_leaf': 2,

'n_train'      : len(X_train),

'n_test'       : len(X_test),

})])

metrics_path = os.path.join(out_dir, 'model_metrics.csv')

metrics_df.to_csv(metrics_path, index=False)

print(f'✓ Metrics CSV      → {metrics_path}')


print("\n" + "=" * 55)

print("✓ ALL DONE — files ready for React dashboard:")

print("=" * 55)

print(f' {raster_path}')

print(f' {model_path}')

print(f' {eval_path}')

print(f' {mdi_path}')

print(f' {shap_path}')

print(f' {metrics_path}')

```

GOOGLE EARTH ENGINE SCRIPT

```
//
```

```
// FLOOD PREDICTION MODEL — BAYELSA STATE, NIGERIA
```

```
// Feature Extraction Script for Google Earth Engine

//

// SAR Methodology:

// - Speckle filter: Lee 5×5 (Adiri et al., 2017; Dar es Salaam study,
//   Taylor & Francis 2023)

// - Threshold: Otsu automatic method (Otsu, 1979; validated at 96.2%
//   accuracy for SAR flood mapping, Academia.edu 2021)

// - Histogram constrained to -28 to -10 dB water backscatter range
//   (Frontiers in Remote Sensing, 2022)

// - Absolute backscatter guard: VV < -17.5 dB (Nature Communications,
//   2025 — 10-year global SAR flood dataset)

// - Post-classification: JRC permanent water mask, slope > 5% mask,
//   pixel connectivity filter (UN-SPIDER Recommended Practice, GEE)

//

// CRS: WGS 84 / UTM Zone 32N (EPSG:32632)

// Bayelsa spans 5.4°E–6.7°E, falls entirely within the 6°E–12°E band

// UTM Zone 32N is the correct metric projection for distance calculations

//
```

//	—	0.	HELPER	FUNCTIONS
----	---	----	--------	-----------

```

// Lee speckle filter (5×5 window)

// Source: Adiri et al. 2017; Lee 1980; standard for SAR preprocessing
function leeFilter(image) {

  var bandNames = image.bandNames();

  var filtered = bandNames.iterate(function(band, acc) {

    band      = ee.String(band);

    var img    = image.select(band);

    var mean   = img.reduceNeighborhood(ee.Reducer.mean(), ee.Kernel.square(2));

    var variance = img.reduceNeighborhood(ee.Reducer.variance(), ee.Kernel.square(2));

    var imgVar  = variance.subtract(mean.pow(2).multiply(0.1));

    var weight  = imgVar.divide(variance.add(1e-10));

    var lee     = mean.add(weight.multiply(img.subtract(mean)));

    return ee.Image(acc).addBands(lee.rename(band));

  }, ee.Image([]));

  return ee.Image(filtered).rename(bandNames);

}

```

```

// Otsu automatic threshold — vectorised via accum(), no ee.List loop

// Source: Otsu (1979); validated at 96.2% accuracy for SAR flood mapping

// Histogram must be pre-constrained to water backscatter range before calling
function otsuThreshold(histogram) {

  var counts = ee.Array(ee.Dictionary(histogram).get('histogram'));

  var means  = ee.Array(ee.Dictionary(histogram).get('bucketMeans'));

  var total  = counts.reduce(ee.Reducer.sum(), [0]).get([0]);

  var sum    = means.multiply(counts).reduce(ee.Reducer.sum(), [0]).get([0]);

```

```

var mean = sum.divide(total);

// Cumulative sums left to right
var aCount = counts.accum(0);
var aSum = means.multiply(counts).accum(0);
var aMean = aSum.divide(aCount.add(1e-10));

// Right-side counts and sums — broadcast scalar to array shape
var totalArray = counts.multiply(0).add(total);
var sumArray = counts.multiply(0).add(sum);
var bCount = totalArray.subtract(aCount);
var bSum = sumArray.subtract(aSum);
var bMean = bSum.divide(bCount.add(1e-10));

// Between-class variance at each candidate threshold
var bss = aCount.multiply(aMean.subtract(mean).pow(2))
    .add(bCount.multiply(bMean.subtract(mean).pow(2)));

// Safe extraction: ee.Array → ee.List → scalar index
var maxIndex = ee.Number(ee.Array(bss.argmax()).toList().get(0)).toInt();
return ee.Number(ee.Array(means).toList().get(maxIndex));
}

```

// — 1. DEFINE BAYELSA BOUNDARY

```
var nigeria = ee.FeatureCollection('FAO/GAUL/2015/level1');  
var bayelsa = nigeria.filter(ee.Filter.eq('ADM1_NAME', 'Bayelsa'));  
var bbox = bayelsa.geometry().bounds();  
  
Map.centerObject(bayelsa, 8);  
Map.addLayer(bayelsa, {color: 'red'}, 'Bayelsa boundary');  
print('Bayelsa area (km²):', bayelsa.geometry().area().divide(1e6).round());
```

// — 2. CHIRPS RAINFALL FEATURES —

```
// Source: CHIRPS v2 — Funk et al. (2015), Scientific Data  
// Validated as lowest RMSE rainfall product in Nigeria  
// (Salami et al., Taylor & Francis 2025)
```

```
var chirpsDaily = ee.ImageCollection('UCSB-CHG/CHIRPS/DAILY')  
  .filterDate('2010-01-01', '2024-12-31')  
  .filterBounds(bayelsa);  
  
// Mean annual rainfall (mm/year)  
var meanRainfall = chirpsDaily.mean()  
  .multiply(365)  
  .rename('mean_annual_rainfall')  
  .clip(bayelsa);
```



```

// Max rainfall intensity — peak daily value across the period

var maxRainfall = chirpsDaily.max()

  .rename('max_rainfall_intensity')

  .clip(bayelsa);


// Rainfall anomaly — flood season vs long-term mean

// Flood season in Bayelsa: August–October (confirmed in literature)

var ltMean = chirpsDaily.mean();

var floodSeasonMean = chirpsDaily

  .filter(ee.Filter.calendarRange(8, 10, 'month'))

  .mean();

var rainfallAnomaly = floodSeasonMean

  .subtract(ltMean)

  .rename('rainfall_anomaly')

  .clip(bayelsa);


Map.addLayer(meanRainfall, {
  min: 2000, max: 4000,
  palette: ['white', 'lightblue', 'blue', 'darkblue']
}, 'Mean Annual Rainfall (mm));

```

// — 3. SRTM TERRAIN FEATURES

// Source: USGS SRTMGL1 — 30m resolution

```
var srtm = ee.Image('USGS/SRTMGL1_003').clip(bayelsa);
```

```
var elevation = srtm.rename('elevation');
```

```
var slope = ee.Terrain.slope(srtm).rename('slope');
```

```
var aspect = ee.Terrain.aspect(srtm).rename('aspect');
```

// Topographic Wetness Index: $TWI = \ln(\text{flow_accum} / \tan(\text{slope}))$

// High TWI = flat, waterlogged terrain — strong flood predictor for Bayelsa

```
var slopeRad = slope.multiply(Math.PI / 180);
```

```
var tanSlope = slopeRad.tan().max(0.001);
```

```
var flowAccum = ee.Image('WWF/HydroSHEDS/15ACC').clip(bayelsa);
```

```
var twi = flowAccum.divide(tanSlope).log().rename('twi');
```

```
Map.addLayer(elevation, {min: 0, max: 50, palette: ['green','yellow','brown']}, 'Elevation (m)');
```

```
Map.addLayer(slope, {min: 0, max: 10, palette: ['white','orange','red']}, 'Slope (°)');
```

```
Map.addLayer(twi, {min: 0, max: 20, palette: ['white','cyan','darkblue']}, 'TWI');
```

// — 4. SENTINEL-1 SAR FLOOD LABELS —————

// Methodology:

// (a) Lee 5×5 speckle filter (Adiri et al. 2017; Taylor & Francis 2023)

// (b) Histogram constrained to -28 to -10 dB water backscatter range

// (Frontiers in Remote Sensing, 2022)

// (c) Otsu auto threshold (Otsu 1979; 96.2% accuracy, PMC 2020)

```
// (d) Absolute guard  $VV < -17.5$  (Nature Communications 2025)  
// (e) Change detection guard (flood image must be below pre-flood)  
// (f) Post-classification (UN-SPIDER Recommended Practice, GEE)
```

```
// 4a. Load SAR collections
```

```
var sarPre = ee.ImageCollection('COPERNICUS/S1_GRD')  
  
  .filterBounds(bayelsa)  
  
  .filterDate('2022-06-01', '2022-07-31') // dry season baseline  
  
  .filter(ee.Filter.eq('instrumentMode', 'IW'))  
  
  .filter(ee.Filter.listContains('transmitterReceiverPolarisation', 'VV'))  
  
  .select('VV')  
  
  .mean()  
  
  .clip(bayelsa);
```

```
var sarFlood = ee.ImageCollection('COPERNICUS/S1_GRD')  
  
  .filterBounds(bayelsa)  
  
  .filterDate('2022-09-01', '2022-10-31') // peak flood season  
  
  .filter(ee.Filter.eq('instrumentMode', 'IW'))  
  
  .filter(ee.Filter.listContains('transmitterReceiverPolarisation', 'VV'))  
  
  .select('VV')  
  
  .mean()  
  
  .clip(bayelsa);
```

```
// 4b. Apply Lee speckle filter to flood image
```

```
var sarFloodFiltered = leeFilter(sarFlood);
```

```
// 4c. Constrain histogram to water backscatter range before Otsu
```

```
// Source: Frontiers in Remote Sensing (2022) — VV range (-28, -10) dB
```

```
var sarMasked = sarFloodFiltered.updateMask(  
  sarFloodFiltered.lt(-10).and(sarFloodFiltered.gt(-28))  
);
```

```
var histogram = sarMasked.reduceRegion({  
  reducer: ee.Reducer.histogram({  
    maxBuckets: 255,  
    minBucketWidth: 0.1  
  }),  
  geometry: bayelsa.geometry(),  
  scale: 500,  
  maxPixels: 1e10  
});
```

```
var otsuValue = otsuThreshold(histogram.get('VV'));  
print('Otsu threshold (dB):', otsuValue);
```

```
// 4d. Apply Otsu + absolute backscatter guard + change detection guard
```

```
var floodRaw = sarFloodFiltered  
  .lt(ee.Image.constant(otsuValue)) // Otsu split  
  .and(sarFloodFiltered.lt(-17.5)) // water backscatter guard (Nature Comms 2025)  
  .and(sarFloodFiltered.lt(sarPre)); // must be lower than pre-flood baseline
```

```
// 4e. Post-classification refinement (UN-SPIDER Recommended Practice)
```

```
// Remove permanent water bodies (present > 10 months/year)
```

```
var jrcSeasonality = ee.Image('JRC/GSW1_4/GlobalSurfaceWater')
```

```
.select('seasonality');
```

```
var permanentWater = jrcSeasonality.gte(10);
```

```
var floodNoPerm = floodRaw.where(permanentWater, 0);
```

```
// Remove steep slopes > 5° — water cannot pool on steep terrain
```

```
var floodNoSteep = floodNoPerm.updateMask(slope.lt(5));
```

```
// Remove isolated pixels connected to 8 or fewer neighbours
```

```
var connected = floodNoSteep.connectedPixelCount(25, true);
```

```
var floodMask = floodNoSteep
```

```
.updateMask(connected.gt(8))
```

```
.rename('flood_label');
```

```
// Diagnostics
```

```
print('Otsu threshold (dB):', otsuValue);
```

```
print('Flood pixel count after refinement:',
```

```
floodMask.reduceRegion({
```

```
  reducer: ee.Reducer.count(),
```

```
  geometry: bayelsa.geometry(),
```

```
  scale: 500,
```

```
maxPixels: 1e10
```

```
  })
```

```
);
```

```
Map.addLayer(sarFloodFiltered, {min: -25, max: 0}, 'SAR flood (Lee filtered)');
```

```
Map.addLayer(floodMask.selfMask(), {palette: ['0000FF']}, 'Flood labels (Otsu)');
```

```
// — 5. LAND USE AND NDVI
```

```
var landcover = ee.Image('MODIS/006/MCD12Q1/2020_01_01')
```

```
  .select('LC_Type1')
```

```
  .clip(bayelsa)
```

```
  .rename('landcover');
```

```
var landsat = ee.ImageCollection('LANDSAT/LC08/C02/T1_L2')
```

```
  .filterBounds(bayelsa)
```

```
  .filterDate('2020-01-01', '2023-12-31')
```

```
  .filter(ee.Filter.lt('CLOUD_COVER', 20))
```

```
  .median();
```

```
var ndvi = landsat
```

```
  .normalizedDifference(['SR_B5', 'SR_B4'])
```

```
  .rename('ndvi')
```

```
  .clip(bayelsa);
```

```
Map.addLayer(ndvi, {  
  min: -0.2, max: 0.8,  
  palette: ['brown', 'white', 'green']  
}, 'NDVI');
```

```
// — 6. DISTANCE TO PERMANENT WATER (JRC-based) —————
```

```
// JRC Global Surface Water occurrence > 50% — more complete than
```

```
// HydroSHEDS for Niger Delta tidal channels and seasonal water bodies
```

```
// fastDistanceTransform outputs true metres under UTM Zone 32N (EPSG:32632)
```

```
var waterOccurrence = ee.Image('JRC/GSW1_4/GlobalSurfaceWater')  
  .select('occurrence')  
  .clip(bayelsa);
```

```
var waterMask = waterOccurrence.gt(50).selfMask();
```

```
var distToRiver = waterMask  
  .fastDistanceTransform(512)  
  .sqrt()  
  .multiply(ee.Image.pixelArea().sqrt())  
  .rename('dist_to_river')  
  .clip(bayelsa);
```

```
Map.addLayer(distToRiver, {  
  min: 0, max: 10000,  
  palette: ['darkblue', 'blue', 'lightblue', 'white']  
}, 'Distance to water (m));
```

```
// — 7. STACK ALL FEATURES — cast to Float32 —————
```

```
// Band order (remember this for Python):
```

```
// 0 elevation
```

```
// 1 slope
```

```
// 2 aspect
```

```
// 3 twi
```

```
// 4 mean_annual_rainfall
```

```
// 5 max_rainfall_intensity
```

```
// 6 rainfall_anomaly
```

```
// 7 ndvi
```

```
// 8 landcover
```

```
// 9 dist_to_river
```

```
// 10 flood_label
```

```
var featureStack = elevation.toFloat()
```

```
.addBands(slope.toFloat())
```

```
.addBands(aspect.toFloat())
```

```
.addBands(twi.toFloat())
```

```
.addBands(meanRainfall.toFloat())
```



```
.addBands(maxRainfall.toFloat())  
.addBands(rainfallAnomaly.toFloat())  
.addBands(ndvi.toFloat())  
.addBands(landcover.toFloat())  
.addBands(distToRiver.toFloat())  
.addBands(floodMask.toFloat());
```

```
print('Feature stack bands:', featureStack.bandNames());
```

```
// — 8. EXPORT TO GOOGLE DRIVE
```

```
// CRS: WGS 84 / UTM Zone 32N (EPSG:32632)  
// Correct for Bayelsa State (5.4°E–6.7°E), within the 6°E–12°E UTM band  
// Scale 500m gives ~600 pixels across the state — good for portfolio model
```

```
Export.image.toDrive({  
  image: featureStack,  
  description: 'Bayelsa_flood_features_v2',  
  folder: 'FloodProject',  
  fileNamePrefix: 'bayelsa_features_v2',  
  region: bbox,  
  scale: 500,  
  crs: 'EPSG:32632',  
  maxPixels: 1e10
```

```
});
```

```
print('✓ Export task created — go to Tasks tab and click RUN.');
```